

SocialTrack Attendance App Documentation

Version 1.0

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For more information, visit

<https://github.com/suneelprojects/Attendance-App.git>

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1 Overview

The SocialTrack Attendance App is a web-based application designed to manage attendance for educational institutions, specifically for administrators and students. Built using React with Vite, Tailwind CSS, and Firebase, it provides a robust system for tracking attendance, managing batches, and generating reports. The application supports both admin and student roles with distinct interfaces and functionalities.

2 Features

- Admin Dashboard: Manage courses, batches, schedules, and attendance records.
- Student Portal: View attendance history, mark attendance, and register for courses.
- Role-Based Access: Secure access using Firebase Authentication with private routes.
- Real-Time Updates: Attendance and schedule data synced with Firebase Firestore.
- Excel Export: Export attendance data for reporting.
- Responsive Design: Built with Tailwind CSS for a mobile-friendly interface.
- Automated Attendance: Auto-mark absent students based on schedules.

3 Tech Stack

- Frontend: React, Vite, Tailwind CSS, Redux Toolkit
- Backend: Firebase (Authentication, Firestore, Hosting)
- Linting: ESLint
- Build Tools: Vite, PostCSS
- Others: Firebase Rules for security, Redux for state management

4 Directory Structure

bharathdatti-socialtrack-attendance-app/	
README.md	% Project overview
AndroidApp.md	% Android app documentation
config.json	% App configuration
dist APR29.zip	% Build archive
eslint.config.js	% ESLint configuration
firestore.rules	% Firebase Firestore security rules
index.html	% Main HTML entry point
LICENSE	% License file
package.json	% NPM dependencies and scripts
postcss.config.js	% PostCSS configuration
tailwind.config.js	% Tailwind CSS configuration
vite.config.js	% Vite configuration

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```

public/
  _redirects      % Netlify redirects src/
  App.jsx         % Main React component index.css  % Global styles
  main.jsx        % React entry point
  components/
    admin/                                     % Admin-specific components
      Attendance.jsx
      BatchManagement.jsx
      Courses.jsx
      Dashboard.jsx
      ResetData.jsx
      Schedule.jsx batch/
        BatchForm.jsx
        BatchList.jsx
        StudentForm.jsx
    auth/                                         % Authentication components
      Login.jsx
      PrivateRoute.jsx
    shared/                                     % Reusable components
      AttendanceTimer.jsx
      CustomLoader.jsx
      LoadingSpinner.jsx
      NotFound.jsx
      UnderConstruction.jsx
    student/                                    % Student-specific components
      AttendanceHistory.jsx
      Registration.jsx
      SharedDeviceAttendance.jsx
      StudentAttendance.jsx firebase/
  config.js layouts/                          % Firebase configuration
  AdminLayout.jsx                             % Admin layout wrapper
  StudentLayout.jsx store/                   % Student layout wrapper

  store.js                                     % Redux store setup
  slices/                                     % Redux slices
    attendanceSlice.js authSlice.js
    batchSlice.js dashboardSlice.js
    scheduleSlice.js
  utils/ attendanceValidation.js % Attendance validation logic autoAbsentMarking.js
    % Auto-mark absent logic
    excelExport.js                          % Excel export functionality
    firebaseHelpers.js                     % Firebase utility functions

```

5 Installation

1. Clone the Repository:

```
git clone https://github.com/bharathdatti/socialtrack-attendance-app. cd bharathdatti-socialtrack-attendance-app
```

2. Install Dependencies:

```
npm install
```

3. Set Up Environment Variables: Create a .env file in the root directory and add your Firebase configuration:

```
VITE_FIREBASE_API_KEY=your-api-key  
VITE_FIREBASE_AUTH_DOMAIN=your-auth-domain  
VITE_FIREBASE_PROJECT_ID=your-project-id  
VITE_FIREBASE_STORAGE_BUCKET=your-storage-bucket  
VITE_FIREBASE_MESSAGING_SENDER_ID=your-messaging-sender-id  
VITE_FIREBASE_APP_ID=your-app-id
```

4. Run the Development Server:

```
npm run dev
```

6 Configuration

- Tailwind CSS: Configured in tailwind.config.js for custom themes and plugins.
- Vite: Build and development settings in vite.config.js.
- ESLint: Linting rules defined in eslint.config.js.
- Firestore Rules: Security rules in firestore.rules to protect data access.
- Config JSON: Application-specific settings in config.json.

7 Usage

7.1 Admin Features

- Dashboard: View attendance summaries and key metrics (Dashboard.jsx).
- Batch Management: Create, update, and delete batches (BatchManagement.jsx, BatchForm.jsx, BatchList.jsx).
- Student Management: Add or remove students from batches (StudentForm.jsx).
- Attendance Tracking: Monitor and update attendance records (Attendance.jsx).
- Schedule Management: Set class schedules (Schedule.jsx).

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- Data Reset: Clear attendance data for a fresh start (ResetData.jsx).
- Courses: Manage available courses (Courses.jsx).

7.2 Student Features

- Registration: Students can register for courses (Registration.jsx).
- Attendance Marking: Mark attendance for classes (StudentAttendance.jsx, SharedDeviceAttendance.jsx).
- Attendance History: View past attendance records (AttendanceHistory.jsx).

8 Firebase Setup

1. Create a Firebase Project: Go to the [Firebase Console](#) and create a new project. Enable Authentication (Email/Password) and Firestore Database.
2. AddFirebaseConfig: Copy the Firebase SDK configuration and add it to src/firebase/config.
3. Deploy Firestore Rules: Update firestore.rules with your security rules and deploy using:
`firebase deploy --only firestore:rules`
4. Hosting: Deploy the app to Firebase Hosting:
`npm run build`
`firebase deploy --only hosting`

9 State Management

The app uses Redux Toolkit for state management, with slices for different domains:

- attendanceSlice.js: Manages attendance data.
- authSlice.js: Handles user authentication state.
- batchSlice.js: Manages batch-related data.
- dashboardSlice.js: Stores dashboard metrics.
- scheduleSlice.js: Tracks class schedules.

Configure the store in store.js and integrate it in main.jsx.

10 Utilities

- attendanceValidation.js: Validates attendance inputs.
- autoAbsentMarking.js: Automatically marks absent students based on schedules.

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- excelExport.js: Exports attendance data to Excel format.
- firebaseHelpers.js: Helper functions for Firebase operations.

11 Deployment

1. Build the App:

```
npm run build
```

2. Deploy to Firebase:

```
firebase deploy
```

3. Netlify Redirects: The `_redirects` file in the public folder handles client-side routing for Netlify deployments.

12 Contributing

1. Fork the repository.
2. Create a feature branch (`git checkout -b feature/your-feature`).
3. Commit changes (`git commit -m 'Add your feature'`).
4. Push to the branch (`git push origin feature/your-feature`).
5. Open a pull request.

13 License

This project is licensed under the MIT License. See the LICENSE file for details.

